Water in the weather: which change of state?

Clouds

condensation

vapour cools high up and forms tiny drops



Rain

no change

drops in clouds grow and fall

Mist / fog

condensation

a cloud close to the ground



Snow

freezing / desublimation

tiny ice crystals join into snowflakes



Hail

freezing

drops freeze while winds push them up and down



Dew

condensation

vapour turns into drops on cold surfaces at night



Frost

desublimation

vapour turns directly into ice on very cold surfaces



Contrails

desublimation

vapour from engines turns into ice crystals

All of them are **physical changes**: the substance is always water, H_2O.